CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

AI engineer (MSc in Artificial Intelligence, 3 years of experience) who builds, tests and deploys AI agents end to end and works directly with the people who use them. Sole developer at an early-stage start-up, building an LLM agent system tested against SAP (ERP) data with SAP domain experts (precision/recall 33%/45% → 93%/90%); full-stack apps in Python and React, with Postgres, deployed to AWS and Azure; AI consulting internship with business stakeholders at Covestro.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer at an early-stage start-up, owning an LLM agent system for the German energy sector end to end (data ingestion, chunking, prompt engineering, indexing), working with SAP domain experts and reporting directly to the CTO.
- Built multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents in parallel) and tested it against ground truth derived from SAP (ERP) system data, raising precision/recall from 33%/45% to 93%/90%.
- Developed Expert Interface, a full-stack web app (React frontend, Python microservices backend) deployed on Azure where domain experts review and approve knowledge-graph changes produced by the agents.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Built a Model Context Protocol (MCP) server exposing the knowledge graph with millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- **Tech stack: Python, LLMs (LiteLLM), LangGraph, LangChain, Postgres, Neo4j, MongoDB, FastAPI, React, MCP, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Advised business stakeholders on how agentic AI could support their decision-making workflows and translated their needs into a solution approach and implementation.
- Built and deployed the full stack: LangGraph agent routing requests across vector database queries, general inference, and summarization; Python/FastAPI backend; React frontend; API testing with Postman.
- Containerized with Docker, deployed to AWS cloud services with CDK, and version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, React, LangGraph, LangChain, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated testing of a RAG assistant (generation and retrieval) with RAGAS and PytrecEval against chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS infrastructure (Lambda, SQS, DynamoDB) with CDK for PrismaChat RAG evaluation.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Order Inbox Agent | Python | Django | TypeScript | Postgres | LLMs

- AI agent for order processing that reads purchase orders from email bodies, PDFs, and Excel files, extracts customer, items, and quantities, matches each line to a product catalog (embeddings plus fuzzy matching), and creates a sales order through a REST API integration with a mock ERP.
- Django backend with Postgres and a TypeScript review screen where a sales rep confirms low-confidence matches before the order is sent; per-customer configuration and a labelled test set of orders run in CI.

TECHNICAL SKILLSET

- **AI Agents & LLMs:** LLM APIs (LiteLLM), LangGraph, LangChain, multi-agent systems, prompt engineering, RAG, embeddings, MCP, evaluation (RAGAS, PytrecEval), Claude Code
- **Full-Stack:** Python, FastAPI, REST APIs, React (beginner), Streamlit, Postman
- **Data:** Postgres, SQL, MongoDB, Neo4j, DynamoDB, vector databases
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, SQS, S3), Azure, Docker, Kubernetes, Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)